



Minimum entropy of a log-concave variable with fixed variance

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Abstract

We show that for log-concave real random variables with fixed variance the Shannon differential entropy is minimized for an exponential random variable, answering a 2010 question of Bobkov and Madiman [1]. This gives a sharp reversal of the celebrated entropy maximization theorem due to Boltzmann, in the log-concave case. We apply this result to derive upper bounds on capacities of additive noise channels with log-concave noise. We also improve constants in the reverse entropy power inequalities for log-concave random variables.

Keywords entropy · variance · log-concave random variables

Mathematics Subject Classification 94A17 · 60E15

1 Introduction

For a real random variable X with density f its differential entropy is defined via the formula $h(X) = h(f) = -\int f \log f$. This definition goes back to the celebrated work of Shannon [2], but the same quantity was also considered, without the minus sign, by physicists, including Boltzmann, in the context of thermodynamics of gases.

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In fact, it is a classical theorem going back to Boltzmann [3] that under fixed variance the entropy is maximized for a Gaussian random variable. This leads to the translation and scale invariant inequality

$$h(X) \leq \frac{1}{2} \log \text{Var}(X) + \frac{1}{2} \log(2\pi e).$$

One can see that in general one cannot hope for a reverse bound, since for the density $f_\varepsilon(x) = (2\varepsilon)^{-1} \mathbf{1}_{[1, 1+\varepsilon]}(|x|)$ the variance remains bounded while the entropy escapes to $-\infty$ as $\varepsilon \rightarrow 0^+$. However, reverse bounds do hold under further assumptions, like log-concavity on X . Recall that X is said to be log-concave if its density is of the form $f = e^{-V}$, where $V: \mathbb{R} \rightarrow (-\infty, \infty]$ is convex.

In [1] the authors showed that indeed, in this class Boltzmann's inequality can be reversed, proving $h(X) \geq \frac{1}{2} \log \text{Var}(X) + \log C$, where C can be taken as $\frac{1}{\sqrt{2}}$ and suggested that an optimal such C , as well as extremizing distributions, should be identified. This constant was improved in [4] where it was shown that the inequality holds with $C = 2$. The goal of this article is to resolve the question of Bobkov and Madiman [1] and prove the following optimal inequality.

Theorem 1.1 *For a log-concave random variable X we have*

$$h(X) \geq \frac{1}{2} \log \text{Var}(X) + 1$$

with equality for the standard one-sided exponential random variable with density $e^{-x} \mathbf{1}_{[0, \infty)}(x)$.

In fact we will show that under fixed variance, the exponential is an extremizer for the class of Rényi entropies of order $\alpha > 1$. Recall that the Rényi entropy of order $\alpha \in (0, \infty) \setminus \{1\}$ [5] is defined as

$$h_\alpha(X) = h_\alpha(f) = \frac{1}{1-\alpha} \log \left(\int f^\alpha(x) dx \right) = -\log \|f(X)\|_{\alpha-1},$$

assuming that the integral converges, see [6] for more background. Under conditions much milder than log-concavity, with $\alpha \rightarrow 1$ one recovers¹ the usual Shannon differential entropy $h(f) = h_1(f) = -\int f \ln f$. Also, by taking limits one can put $h_0(f) = \log |\text{Supp } f|$, where $\text{Supp } f$ stands for the support of f and $h_\infty(f) = -\log \|f\|_\infty$, where $\|f\|_\infty$ is the essential supremum of f . For $p \geq q > 0$ one has

$$0 \leq h_q(f) - h_p(f) \leq \frac{\log q}{q-1} - \frac{\log p}{p-1},$$

where the second inequality holds thanks to the log-concavity of f , with equality for the standard one-sided exponential density, see [9]. Here the fraction $\frac{\log t}{t-1}$ is interpreted

¹ For example, if $\int f^\beta(x) dx < \infty$ for some $\beta > 1$, $\lim_{\alpha \rightarrow 1} h_\alpha(X) = h(X)$. We refer the reader to [7, Lemma 5.1] and [8, Lemma 2] for more on these technicalities.

as 1 for $t = 1$. Thus using this bound together with Theorem 1.1 gives the following corollary.

Corollary 1.2 *For $\alpha > 1$ and a log-concave random variable X one has*

$$h_\alpha(X) \geq \frac{1}{2} \log \text{Var}(X) + \frac{\log \alpha}{\alpha - 1},$$

with equality when X is an exponential random variable.

Let us mention that the problem of minimizing Rényi entropy under fixed variance for *symmetric* log-concave random variables was solved in [10] for the case $\alpha \leq 1$ and in [11] for $\alpha > 1$. Here the worst case is the uniform random variable on an interval for $\alpha \leq \alpha^*$ and the symmetric exponential random variable for $\alpha > \alpha^*$, where $\alpha^* \approx 1.241$ is the solution to the equation $\frac{\log \alpha^*}{\alpha^* - 1} = \frac{1}{2} \log 6$.

2 Applications

2.1 Additive noise channels

We now briefly discuss an application of our main result in the context of information theory. For more details we refer the reader to [10], where the case of symmetric log-concave random variables was discussed. Consider the memoryless transmission channel with power budget P subject to additive noise N , that is, if the input of the channel is X , then output produced by the channel is $Y = X + N$, where N is the noise independent of X . Shannon's celebrated channel coding theorem [2] asserts that the so-called *capacity* of such a channel is given by the formula

$$C_P(N) = \sup_{X: \text{Var}(X) \leq P} (h(X + N) - h(N)).$$

We have the following fact, see [2].

Proposition 2.1 *Let N be a random variable with finite variance and let Z be a centered Gaussian random variable with the same variance. Then*

$$C_P(Z) \leq C_P(N) \leq C_P(Z) + D(N),$$

where $D(N) = h(Z) - h(N)$ is the relative entropy of N from Gaussianity.

Our Theorem 1.1 gives

$$D(N) = h(Z) - h(N) \leq h(Z) - \frac{1}{2} \log \text{Var}(N) - 1 = \frac{1}{2} \log \left(\frac{2\pi}{e} \right).$$

We can therefore formulate the following corollary.

Corollary 2.2 *Let N be a log-concave noise and let Z be a centered Gaussian noise with the same variance. Then*

$$C_P(Z) \leq C_P(N) \leq C_P(Z) + \frac{1}{2} \log \left(\frac{2\pi}{e} \right).$$

In other words, using an arbitrary log-concave noise instead of the Gaussian does not increase capacity by more than $\frac{1}{2} \log \left(\frac{2\pi}{e} \right) < 0.42$ nats.

2.2 Reverse EPI

Recall that the entropy power of a real random variable X is defined by $\mathcal{N}(X) = \frac{1}{2\pi e} \exp(2h(X))$. Note that for a Gaussian random variable one has $\mathcal{N}(Z) = \text{Var}(Z)$ and in general $\text{Var}(X) = \text{Var}(Z)$ implies $\mathcal{N}(X) \leq \mathcal{N}(Z)$. Theorem 1.1 can be rewritten in the form $\mathcal{N}(X) \geq \frac{e}{2\pi} \text{Var}(X)$. The entropy power inequality of Shannon and Stam [2, 12] states that for independent random variables X, Y one has

$$\mathcal{N}(X + Y) \geq \mathcal{N}(X) + \mathcal{N}(Y).$$

It is of interest to obtain reverse bounds. Under the log-concavity assumption the authors of [4] showed that if X, Y are uncorrelated, then $\mathcal{N}(X + Y) \leq \frac{\pi e}{2} (\mathcal{N}(X) + \mathcal{N}(Y))$. Using Theorem 1.1 we can improve this result.

Corollary 2.3 *Let X, Y be log-concave uncorrelated real random variables. Then*

$$\mathcal{N}(X + Y) \leq \frac{2\pi}{e} (\mathcal{N}(X) + \mathcal{N}(Y)).$$

Indeed, one has

$$\mathcal{N}(X + Y) \leq \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) \leq \frac{2\pi}{e} (\mathcal{N}(X) + \mathcal{N}(Y)).$$

We can also define the Rényi entropy power via $\mathcal{N}_\alpha(X) = \exp(2h_\alpha(X))$, where we removed the normalizing constant $2\pi e$ for simplicity. Similarly as in [11], Theorem 2 from [13] together with our Theorem 1.1 gives the inequality

$$C_-(\alpha) \text{Var}(X) \leq \mathcal{N}_\alpha(X) \leq C_+(\alpha) \text{Var}(X), \quad \alpha > 1,$$

where

$$C_-(\alpha) = \alpha^{\frac{2}{\alpha-1}}, \quad C_+(\alpha) = \frac{3\alpha-1}{\alpha-1} \left(\frac{2\alpha}{3\alpha-1} \right)^{\frac{2}{1-\alpha}} B\left(\frac{1}{2}, \frac{\alpha}{\alpha-1}\right)^2.$$

Here $B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$ stands for the Beta function. The right inequality does not need log-concavity. Thus, using the same computation as for the case $\alpha = 1$ we get the following corollary.

Corollary 2.4 *If X, Y are log-concave uncorrelated real random variables, then for $\alpha > 1$ one has*

$$\mathcal{N}_\alpha(X + Y) \leq \frac{C_+(\alpha)}{C_-(\alpha)} (\mathcal{N}_\alpha(X) + \mathcal{N}_\alpha(Y)).$$

3 Reductions

In this section we reduce the inequality $h(X) \geq \frac{1}{2} \log \text{Var}(X) + 1$ to variables with monotone two-piece affine densities (see below for a precise definition). To achieve this we will make use of two standard ingredients, namely rearrangement together with degrees of freedom/localization techniques.

3.1 Decreasing Rearrangement

For a measurable set $A \subseteq \mathbb{R}$ let $|A|$ denote its Lebesgue measure and let us define $A^\downarrow = (0, |A|) \subseteq \mathbb{R}$ with the interpretation of $(0, 0)$ as the empty set. For a measurable function $f : \mathbb{R} \rightarrow [0, \infty]$, we define $f^\downarrow : (0, \infty) \rightarrow [0, \infty]$ as

$$f^\downarrow(x) = \int_0^\infty \mathbb{1}_{\{y: f(y) > \lambda\}^\downarrow}(x) d\lambda.$$

We make some elementary observations, starting with the fact that $f^\downarrow$ is fully characterized by the equality $\{f > \lambda\}^\downarrow = \{f^\downarrow > \lambda\}$ and in particular, equimeasurable (with respect to the Lebesgue measure) functions possess identical decreasing rearrangements. The main result of this subsection is Theorem 3.3, which reduces our problem to decreasing densities.

Proposition 3.1 *For $f : \mathbb{R} \rightarrow [0, \infty)$, its rearrangement $f^\downarrow$ satisfies,*

$$f^\downarrow(x) = \sup\{\lambda : x \in \{f > \lambda\}^\downarrow\} \quad (1)$$

and

$$\{f^\downarrow > \lambda\} = \{f > \lambda\}^\downarrow. \quad (2)$$

Proof Representation (1) follows directly from the definition of $f^\downarrow$. To prove (2) observe that by (1) the condition $f^\downarrow(x) > \lambda$ is equivalent to the existence of $\lambda' > \lambda$ such that $x \in \{f > \lambda'\}^\downarrow$. It is therefore enough to prove the following equivalence

$$\exists \lambda' > \lambda \quad x \in \{f > \lambda'\}^\downarrow \iff x \in \{f > \lambda\}^\downarrow.$$

By nestedness the implication $\implies$ is trivial. To show the other direction assume that $x \in \{f > \lambda\}^\downarrow$. This is equivalent to $x < |\{f > \lambda\}|$. By the continuity of Lebesgue

measure

$$x < |\{f > \lambda\}| = \left| \bigcup_n \left\{ f > \lambda + \frac{1}{n} \right\} \right| = \lim_n |\{f > \lambda + 1/n\}|.$$

By taking $\lambda' = \lambda + \frac{1}{n}$ for large enough n we get $\lambda' > \lambda$ such that $x \in \{f > \lambda'\}^\downarrow$. $\square$

Proposition 3.2 For ϕ and f Borel measurable with f non-negative,

$$\int \phi(f(x))dx = \int \phi(f^\downarrow(x))dx.$$

Proof This is equivalent to the statement that f pushes the Lebesgue measure dm forward to the same measure $f^\downarrow$ pushes the Lebesgue measure to. Thus it suffices to check

$$f_\# m(\lambda, \infty) = f^\downarrow_\# m(\lambda, \infty).$$

This is just $|\{f > \lambda\}| = |\{f^\downarrow > \lambda\}|$, which follows from the characterization $\{f^\downarrow > \lambda\} = \{f > \lambda\}^\downarrow$. $\square$

Note that taking $\phi(x) = -x \log x$ shows that the decreasing rearrangement $f^\downarrow$ preserves the entropy of a density function f .

For a random variable X with density f , define $X^\downarrow$ to be a random variable drawn from the density $f^\downarrow$. When X has density f we will write the variance as $\text{Var}(X)$ and $\text{Var}(f)$ interchangeably.

Theorem 3.3 For a log-concave real random variable X we have

$$\text{Var}(X) \leq \text{Var}(X^\downarrow).$$

Proof Without loss of generality suppose that f is supported on $[0, \infty)$ and let

$$a(\lambda) := \inf\{x : f(x) \geq \lambda\}, \quad b(\lambda) := \sup\{x : f(x) \geq \lambda\}$$

and $d\theta(\lambda) = (b(\lambda) - a(\lambda))d\lambda$, where $d\lambda$ is the Lebesgue measure on $[0, \infty)$. Then

$$f(x) = \int_0^\infty \frac{\mathbb{1}_{[a(\lambda), b(\lambda)]}(x)}{b(\lambda) - a(\lambda)} d\theta(\lambda),$$

where a is non-decreasing and b is non-increasing. We have $\int d\theta = 1$ with θ supported on $[0, \|f\|_\infty]$. Then

$$f^\downarrow(x) = \int_0^\infty \frac{\mathbb{1}_{[0, b(\lambda) - a(\lambda)]}(x)}{b(\lambda) - a(\lambda)} d\theta(\lambda)$$

and thus for positive integers p we have

$$\int x^p f(x)dx = \frac{1}{p+1} \int_0^\infty \frac{b^{p+1} - a^{p+1}}{b - a} d\theta$$

and

$$\int x^p f^\downarrow(x) dx = \frac{1}{p+1} \int_0^\infty (b-a)^p d\theta.$$

Since our inequality can be rewritten as

$$\int x^2(f(x) - f^\downarrow(x))dx \leq \left(\int x(f(x) - f^\downarrow(x))dx \right) \left(\int x(f(x) + f^\downarrow(x))dx \right),$$

our goal is to show that

$$\begin{aligned} & \frac{1}{3} \int_0^\infty (b^2 + ba + a^2 - (b-a)^2) d\theta \\ & \leq \frac{1}{4} \left(\int_0^\infty (b+a - (b-a)) d\theta \right) \left(\int_0^\infty (b+a + (b-a)) d\theta \right). \end{aligned}$$

This reduces to

$$\int ab d\theta \leq \left(\int a d\theta \right) \left(\int b d\theta \right),$$

which is the so-called Chebyshev's other inequality. This can be proved by observing that

$$\begin{aligned} & 2 \int_0^\infty ab d\theta - 2 \left(\int_0^\infty a d\theta \right) \left(\int_0^\infty b d\theta \right) \\ & = \int_0^\infty \int_0^\infty (a(\lambda_1) - a(\lambda_2))(b(\lambda_1) - b(\lambda_2)) d\theta(\lambda_1) d\theta(\lambda_2) \leq 0 \end{aligned}$$

since $(a(\lambda_1) - a(\lambda_2))(b(\lambda_1) - b(\lambda_2)) \leq 0$ for all λ_1, λ_2 as a is non-decreasing and b is non-increasing. $\square$

Lemma 3.4 *For a log-concave real random variable X we have*

$$\frac{e^{2h(X)}}{\text{Var}(X)} \geq \frac{e^{2h(X^\downarrow)}}{\text{Var}(X^\downarrow)}.$$

This follows immediately from Theorem 3.3 and Proposition 3.2 (with $\phi(x) = -x \log x$) since decreasing rearrangement will preserve entropy and increase variance. It allows us to reduce our problem to X log-concave with monotone decreasing density, in light of the following.

Proposition 3.5 *For X log-concave, $X^\downarrow$ is log-concave as well.*

This is an immediate corollary of a more general result, Proposition 7.5.8 in [14]. We include the proof here for the convenience of the reader.

Proof Note that X has a log-concave density f if and only if

$$(1-t)\{f > \lambda_1\} + t\{f > \lambda_2\} \subseteq \{f > \lambda_1^{1-t} \lambda_2^t\} \quad (3)$$

for all $t \in (0, 1)$ and $\lambda_1, \lambda_2 > 0$. Thus, to show that $X^\downarrow$ is log-concave it suffices to prove

$$(1-t)\{f > \lambda_1\}^\downarrow + t\{f > \lambda_2\}^\downarrow \subseteq \{f > \lambda_1^{1-t}\lambda_2^t\}^\downarrow.$$

But since both sets are open intervals with the left-endpoints at 0 it suffices to prove that the right hand side has bigger volume, which follows from the translation invariance of the Lebesgue measure and the set theoretic inclusion in (3),

$$\begin{aligned} |(1-t)\{f > \lambda_1\}^\downarrow + t\{f > \lambda_2\}^\downarrow| &= |(1-t)\{f > \lambda_1\} + t\{f > \lambda_2\}| \\ &\leq |\{f > \lambda_1^{1-t}\lambda_2^t\}| = |\{f > \lambda_1^{1-t}\lambda_2^t\}^\downarrow|. \end{aligned}$$

□

3.2 Degrees of freedom

Our goal is to show that in order to prove Theorem 1.1 it suffices to consider functions of the form e^{-V} where V is two-piece affine on a finite interval. This is a standard argument appearing e.g. in [10], so we only sketch it.

Step 1. Let $\sigma^2 = \text{Var}(X)$. By the previous subsection, in order to prove the inequality $h(f) \geq \log(e\sigma)$ it suffices to consider non-increasing densities on $[0, \infty)$. In fact by an approximation argument we can assume that the support of f is a finite interval. Indeed if we define $f_n = e^{-V} \mathbb{1}_{(0,n)}/c_n$, where $c_n = \int_0^n e^{-V}$ is the normalizing constant, then clearly $c_n \rightarrow 1$ as $n \rightarrow \infty$ and by the Lebesgue dominated convergence theorem $\text{Var}(f_n) \rightarrow \text{Var}(f)$. We also have

$$h(f_n) = \frac{1}{c_n} \int_0^n V e^{-V} + \frac{\log c_n}{c_n} \int_0^n e^{-V} \rightarrow h(f)$$

by the Lebesgue dominated convergence theorem, as

$$|V|e^{-V} = |V|e^{-V/2}e^{-V/2} \leq C e^{-V/2},$$

where $C = \max_{t \geq V(0)} |t|e^{-t/2}$ (the function V is convex and thus increases at least linearly at infinity, so $e^{-V/2}$ is integrable).

Step 2. We can therefore fix the interval $[0, L]$ on which the function f is defined. Let $A = A_{L,\sigma}$ denote the set of log-concave non-increasing densities supported in $[0, L]$ and having variance σ^2 . We have $f(0)^2 \text{Var}(X) \leq f(0)^2 \mathbb{E}X^2 \leq 2$, see [15]. Let us give a simple proof following the argument from [10]. By scaling we can assume that $f(0) = 1$. If $g(x) = e^{-x} \mathbb{1}_{[0,\infty)}(x)$ then by the log-concavity of f the function $f - g$ changes sign in exactly one point $a > 0$ and thus

$$\mathbb{E}X^2 - 2 = \int_0^\infty x^2(f(x) - g(x))dx = \int_0^\infty (x^2 - a^2)(f(x) - g(x))dx \leq 0$$

since the integrand is non-positive. This shows that f is bounded by $2^{\frac{1}{2}}\sigma^{-1}$ and in particular $h(f) = -\int f \log f \geq -\log f(0)$ is bounded from below. This shows that

the quantity $M = \inf\{h(f) : f \in A_{L,\sigma}\}$ is finite. Let f_n be such that $h(f_n) \rightarrow M$. By a straightforward adaptation of Lemma 12 from [10] we get that (f_n) has an almost everywhere convergent subsequence $f_{n_k} \rightarrow f_0 \in A$ and by the Lebesgue dominated convergence theorem $h(f_0) = M$. This shows that the infimum of $h(f)$ is attained on A .

Step 3. We now apply the theory of degrees of freedom due to Fradelizi and Guédon from [16]. We say that $f \in A$ has d degrees of freedom if there exist $\varepsilon > 0$ and linearly independent functions $f_1, \dots, f_d$ such that $f_\delta := f + \sum_{i=1}^d \delta_i f_i$ is log-concave and non-increasing for all $|\delta_i| \leq \varepsilon$. Suppose f has at least 4 degrees of freedom. Then

$$\int f_\delta(x) dx = \int f(x) dx, \quad \int x f_\delta(x) dx = \int x f(x) dx, \quad \int x^2 f_\delta(x) dx = \int x^2 f(x) dx$$

is a system of 3 linear equations in variables $\delta_1, \dots, \delta_4$ and thus has nontrivial linear subspace of solutions. Thus there is a non-zero $\delta = (\delta_1, \dots, \delta_4)$ such that $f_\delta, f_{-\delta} \in A$ and

$$h(f) = h\left(\frac{1}{2}f_\delta + \frac{1}{2}f_{-\delta}\right) > \frac{1}{2}h(f_\delta) + \frac{1}{2}h(f_{-\delta})$$

since entropy is a strictly convex functional. Thus f is not the extremizer of $h(f)$. This shows that extremizers have to have at most 3 degrees of freedom and Step IV of the proof of Theorem 1 from [10] shows that such functions must be two piece log-affine, that is $f = e^{-V}$ where $V = \max\{\varphi_1, \varphi_2\}$ for φ_i log-affine. The space of log-concave random variables with compactly supported, non-increasing, two piece log-affine densities on $[0, \infty)$ will be denoted by $\mathcal{F}_2$.

3.3 Localization

Alternatively, we can proceed with extreme point analysis following Fradelizi and Guedon [16]. For a fixed compact interval $K \subseteq \mathbb{R}$, and upper semi-continuous functions g_1, g_2 define $\mathcal{P}_g$ to be the space of log-concave probability measures μ supported in K such that

$$\int g_i d\mu \geq 0, \quad i = 1, 2.$$

We will use the following special case of Fradelizi and Guedon's result.

Theorem 3.6 ([16] Theorem 1) *Let ν be an extreme point of the convex hull of $\mathcal{P}_g$, then ν is a point mass, or ν has density e^{-V} with respect to the Lebesgue measure where on the support of ν , $V = \max\{\phi_1, \phi_2\}$ for ϕ_i affine.*

We can now prove the following lemma.

Lemma 3.7 *For a log-concave real random variable X , we have*

$$\frac{e^{2h(X)}}{\text{Var}(X)} \geq \inf_{Z \in \mathcal{F}_2} \frac{e^{2h(Z)}}{\text{Var}(Z)}.$$

Proof Recalling the truncation argument it suffices to consider $X \sim \mu$ with density supported on $[0, L]$ for some $L > 0$. Fix X and take

$$\begin{aligned} g_1(x) &= \mathbb{E}[X] - x \\ g_2(x) &= x^2 - \mathbb{E}[X^2]. \end{aligned}$$

For $Z \sim \nu$ an extreme point of $\mathcal{P}_g$ (here $\mathcal{P}_g$ is defined with $K = [0, L]$, see the definition right before Theorem 3.6), since Z is non-negative, by $\int g_i d\nu \geq 0$ we have

$$\text{Var}(Z) \geq \text{Var}(X) > 0,$$

so ν is not a point mass and hence by Theorem 3.6, Z has a two piece log-affine density. Since $X \sim \mu \in \mathcal{P}_g$ by definition, if we let $\mathcal{E}(\mathcal{P}_g)$ denote the extreme points of the convex hull of $\mathcal{P}_g$, by the Krein-Milman theorem μ belongs to the closure of the convex hull of $\mathcal{E}(\mathcal{P}_g)$. Now let us show that this implies that

$$h(X) \geq \inf_{Z \sim \nu \in \mathcal{E}(\mathcal{P}_g)} h(Z). \quad (4)$$

Indeed, the entropy is concave and upper semi-continuous in the weak topology when restricted to compact sets (as can be seen from the more well known fact that the relative entropy is lower semicontinuous, and on compact sets $h(X) = h(U) - D(X||U)$ where U is the uniform distribution on the compact set), thus writing μ as $\lim_n \mu_n$ for a sequence of μ_n that can be expressed as convex combinations of extreme points, that is $\mu_n = \sum_{i=1}^{k_n} \lambda_n(i) \nu_n(i)$ for $\nu_n(i) \in \mathcal{E}(\mathcal{P}_g)$, we have

$$\begin{aligned} h(\mu) &\geq \limsup_n h(\mu_n) = \limsup_n h\left(\sum_{i=1}^{k_n} \lambda_n(i) \nu_n(i)\right) \\ &\geq \limsup_n \sum_{i=1}^{k_n} \lambda_n(i) h(\nu_n(i)) \geq \inf_{\nu \in \mathcal{E}(\mathcal{P}_g)} h(\nu). \end{aligned}$$

Since every element of $\mathcal{P}_g$ has variance no smaller than X , it follows that

$$\frac{e^{2h(X)}}{\text{Var}(X)} \geq \inf_{Z \sim \nu \in \mathcal{E}(\mathcal{P}_g)} \frac{e^{2h(Z)}}{\text{Var}(Z)}.$$

Considering $Z^\downarrow$ for $Z \in \mathcal{E}(\mathcal{P}_g)$, and applying Lemma 3.4, we have

$$\frac{e^{2h(X)}}{\text{Var}(X)} \geq \inf_{Z \sim \nu \in \mathcal{E}(\mathcal{P}_g)} \frac{e^{2h(Z^\downarrow)}}{\text{Var}(Z^\downarrow)}.$$

Direct computation shows that if Z has a two-piece log-affine density, then $Z^\downarrow$ does as well, completing the proof. $\square$

4 Proof of Theorem 1.1

The proof of Theorem 1.1 goes in 3 steps. The above preliminaries first allow us to reduce to a two-piece affine function which, after re-parametrisation, amounts to proving that a 3 variable function $G(c, x, y)$ is non-negative. As a second technical step, we remove an exponential factor to reduce to a simpler function, that we finally study by hand.

4.1 Three-point inequality

From the previous section we can assume that $f = e^{-V}$, where V is two-piece affine and non-decreasing on an interval $[0, L]$. In fact by scale invariance and translation we can assume that we have the following parametrization of our function

$$g(t) = e^{-\frac{t}{a}} \mathbb{1}_{[-ax, 0]}(t) + e^{-\frac{t}{b}} \mathbb{1}_{[0, -yb]}(t), \quad f = \frac{g}{c},$$

$$c = \int g = a(e^x - 1) - b(e^y - 1),$$

where

$$a \geq b > 0, \quad x \geq 0, \quad y \leq 0.$$

Then f is a probability density and we have

$$\int xg(x)dx = a^2(e^x(1-x) - 1) - b^2(e^y(1-y) - 1)$$

and

$$\int x^2g(x)dx = a^3(e^x(x^2 - 2x + 2) - 2) - b^3(e^y(y^2 - 2y + 2) - 2).$$

Also

$$-\int g \log g = a(e^x(1-x) - 1) - b(e^y(1-y) - 1).$$

The inequality of Theorem 1.1 we want to prove is

$$-\int f \log f \geq \frac{1}{2} \log \text{Var}(f) + 1.$$

This is

$$e^{-2 \int f \log f} \geq e^2 \text{Var}(f).$$

In terms of g

$$e^{-2} e^{-2 \int \frac{g}{c} \log \frac{g}{c}} \geq \frac{1}{c} \int x^2 g(x) dx - \frac{1}{c^2} \left(\int x g(x) dx \right)^2.$$

We have

$$e^{-2} e^{-2 \int \frac{g}{c} \log \frac{g}{c}} = e^{-2} e^{-\frac{2}{c} \int g \log g + 2 \log c} = e^{-2} c^2 e^{-\frac{2}{c} \int g \log g},$$

so after multiplying both sides by c^2 we want to prove

$$e^{-2} c^4 e^{-\frac{2}{c} \int g \log g} \geq c \int x^2 g(x) dx - \left(\int x g(x) dx \right)^2.$$

Observe that

$$\begin{aligned} e^{-2} e^{-\frac{2}{c} \int g \log g} &= \exp \left(2 \frac{a(e^x(1-x) - 1) - b(e^y(1-y) - 1)}{a(e^x - 1) - b(e^y - 1)} - 2 \right) \\ &= \exp \left(2 \frac{-axe^x + bye^y}{a(e^x - 1) - b(e^y - 1)} \right). \end{aligned}$$

Our goal is therefore to prove

$$c^4 \exp \left(2 \frac{-axe^x + bye^y}{a(e^x - 1) - b(e^y - 1)} \right) \geq c \int x^2 g(x) dx - \left(\int x g(x) dx \right)^2.$$

Equivalently

$$\begin{aligned} &(a(e^x - 1) - b(e^y - 1))^4 \exp \left(-2 \frac{ae^x x - be^y y}{a(e^x - 1) - b(e^y - 1)} \right) \geq \\ &(a(e^x - 1) - b(e^y - 1)) \left(a^3 \left(e^x (x^2 - 2x + 2) - 2 \right) - b^3 \left(e^y (y^2 - 2y + 2) - 2 \right) \right) \\ &- \left(a^2 (e^x(1-x) - 1) - b^2 (e^y(1-y) - 1) \right)^2. \end{aligned}$$

The inequality is invariant under $(a, b) \rightarrow (ta, tb)$ and therefore we can assume that $b = 1$ and $a \geq 1$. Let us define the function

$$\begin{aligned} G(c, x, y) &= (c(e^x - 1) - (e^y - 1))^4 \exp \left(-2 \frac{ce^x x - e^y y}{c(e^x - 1) - (e^y - 1)} \right) \\ &- (c(e^x - 1) - (e^y - 1)) \left(c^3 \left(e^x (x^2 - 2x + 2) - 2 \right) - \left(e^y (y^2 - 2y + 2) - 2 \right) \right) \\ &+ \left(c^2 (e^x(1-x) - 1) - (e^y(1-y) - 1) \right)^2. \end{aligned}$$

All together, we reduced the proof of Theorem 1.1 to proving that $G(c, x, y) \geq 0$ for $x \geq 0$, $y \leq 0$ and $c \geq 1$.

4.2 Estimating the exponent

We observe that

$$\frac{ce^x x - e^y y}{c(e^x - 1) - (e^y - 1)} - x = \frac{(x - y)e^y + x(c - 1)}{c(e^x - 1) - (e^y - 1)} \in [0, 1].$$

Indeed, the extremal values of the right hand side are attained for $c = 1, \infty$ and are therefore equal $\frac{(x-y)e^y}{e^x - e^y} = \frac{(x-y)}{e^x - y - 1}$ and $\frac{x}{e^x - 1}$. Since clearly $\frac{\theta}{e^\theta - 1} \in [0, 1]$ for any $\theta > 0$ the claim follows.

Now, on the interval $[0, 2]$ we have the bound $e^{-x} \geq 1 - x + \frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{7}{240}x^4$. The inequality holds for $x = 2$ and thus it is enough to show that the derivative of $g(x) = e^{-x} - (1 - x + \frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{7}{240}x^4)$ is positive or has sign pattern $(+, -)$. It is clear that if $f : \mathbb{R}_+ \rightarrow \mathbb{R}$ satisfies $f(0) = 0$ and f' is positive or has sign pattern $(+, -)$ then f itself is positive or has sign pattern $(+, -)$. Using this argument three times and observing that $g^{(4)}(x) = e^{-x} - \frac{7}{10}$ has sign pattern $(+, -)$ finishes the proof. Using this bound we obtain

$$e^{-2\frac{ce^x x - e^y y}{c(e^x - 1) - (e^y - 1)}} = e^{-L} e^{-2x} \geq e^{-2x} \left(1 - L + \frac{1}{2}L^2 - \frac{1}{6}L^3 + \frac{7}{240}L^4 \right)$$

with $L(c, x, y) = 2\frac{(x-y)e^y + x(c-1)}{c(e^x - 1) - (e^y - 1)}$. This gives a lower bound on $G(c, x, y)$ of the form

$$15e^{2x}G(c, x, y) \geq P_0(x, y) + (c - 1)P_1(x, y) + (c - 1)^2P_2(x, y) + (c - 1)^3P_3(x, y) + (c - 1)^4P_4(x),$$

where $P_0, \dots, P_4$ are explicitly given and studied in the next sections.

To prove Theorem 1.1 it is therefore sufficient to prove that $P_i(x, y) \geq 0$ for $0 \leq i \leq 4$ which we now show.

4.3 Positivity of $P_i(x, y), i = 0, 1, \dots, 4$

To prove that the P_i 's are positive is a tedious but somehow simple exercise. The strategy that we adopt relies on a series expansion in the variable x . For brevity, some details are left to the reader.

We start with the simplest case P_4 that involves only the variable x .

4.3.1 Positivity of $P_4(x)$

We have

$$P_4(x) = 7x^4 + 20x^3 + 30x^2 + 30x + 15 + 15e^{3x}(x^2 - 2x - 2) + 15e^{2x}(2x^2 + 6x + 5) - 10e^x(2x^3 + 6x^2 + 9x + 6).$$

Expanding in x we write $P_4(x) = \sum_{n \geq 0} \frac{a_n}{n!} x^n$, where

$$a_n = -10(2n^3 + 7n + 6) + 15 \cdot 2^{n-1}(n^2 + 5n + 10) + 5 \cdot 3^{n-1}(n-9)(n+2)$$

for $n \geq 5$ and $a_0 = a_1 = a_2 = a_3 = 0$ and $a_4 = 18$. It is easy to check that $a_n \geq 0$ for all $n \geq 0$, as for large n the term $5 \cdot 3^{n-1}n^2$ dominates and for small n the inequality can be checked directly.

4.3.2 Positivity of $P_0(x, y)$

We have

$$\begin{aligned} e^{-y} P_0(x, y) &= 15e^{2x+y} (2x^2 + x(6-4y) + 2y^2 - 6y + 5) \\ &\quad + 15e^{3x} (x^2 - 2x(y+1) + y^2 + 2y - 2) \\ &\quad - 10e^{x+2y} (2x^3 - 6x^2(y-1) + 3x(2y^2 - 4y + 3) - 2y^3 + 6y^2 - 9y + 6) \\ &\quad + e^{3y} (7x^4 - 4x^3(7y-5) + 6x^2(7y^2 - 10y + 5) \\ &\quad + x(-28y^3 + 60y^2 - 60y + 30)) \\ &\quad + e^{3y} (7y^4 - 20y^3 + 30y^2 - 30y + 15). \end{aligned}$$

We Taylor expand the right hand side in the form $\sum_{n \geq 0} \frac{f_n(y)}{n!} x^n$, where

$$f_n(y) = a_0 + a_1 y + a_2 y^2 + e^y(b_0 + b_1 y + b_2 y^2) + e^{2y}(c_0 + c_1 y + c_2 y^2 + c_3 y^3)$$

for $n \geq 5$, with

$$a_0(n) = 3^{n-1}(5n^2 - 35n - 90), \quad a_1(n) = 3^n \cdot 10 \cdot (3-n), \quad a_2(n) = 5 \cdot 3^{n+1}$$

$$b_0(n) = 15 \cdot 2^{n-1}(n^2 + 5n + 10), \quad b_1(n) = -15 \cdot 2^{n+1}(n+3),$$

$$b_2(n) = 15 \cdot 2^{n+1}, \quad c_0(n) = -10(2n^3 + 7n + 6),$$

$$c_1(n) = 30(2n^2 + 2n + 3), \quad c_2(n) = -60(n+1), \quad c_3(n) = 20.$$

The terms with a_1, a_2, b_0, b_1, b_2 are clearly positive. Using consecutively the following crude bounds, valid for any $y \leq 0$, $e^{2y} \leq 1$, $ye^{2y} \geq -1$, $y^2 e^{2y} \leq 1$ and $y^3 e^{2y} \geq -1$ we arrive at $f_n(y) \geq a_0 + c_0 - c_1 + c_2 - c_3$. The leading term in this expression is $5 \cdot 3^{n-1}n^2$ and it is easy to check that the expression is positive for $n \geq 10$.

The inequalities $f_n(y) \geq 0$ for $0 \leq n \leq 9$ can be proved by Taylor expanding $e^{3t} f_n(-t)$ for $0 \leq n \leq 4$ and $e^{2t} f_n(-t)$ for $5 \leq n \leq 9$, and checking that the coefficients are nonnegative. We leave the details to the reader.

4.3.3 Positivity of $P_1(x, y)$

We have

$$\begin{aligned}
 P_1(x, y) = & 15e^{2x+y} \left(4x^2 - 4x(y-3) - y^2 - 8y + 8 \right) \\
 & + 60e^{2(x+y)} \left(x^2 + x(3-2y) + y^2 - 3y + 3 \right) \\
 & + 15e^{3x+y} \left(3x^2 - 4x(y+2) + y^2 + 8y - 8 \right) + 15e^{3x} x^2 \\
 & - 30e^{x+2y} \left(2x^3 + x^2(6-4y) + x(2y^2 - 8y + 9) + 2(y^2 - 3y + 3) \right) \\
 & - 10e^{x+3y} \left(2x^3 - 6x^2(y-1) \right. \\
 & \left. + 3x(2y^2 - 4y + 3) - 2y^3 + 6y^2 - 9y + 6 \right) \\
 & - 2e^{3y} \left(-14x^4 + x^3(42y - 40) - 6x^2(7y^2 - 15y + 10) \right. \\
 & \left. + 2x(7y^3 - 30y^2 + 45y - 30) \right) \\
 & - 10e^{3y} \left(2y^3 - 6y^2 + 9y - 6 \right).
 \end{aligned}$$

We Taylor expand the right hand side in the form $\sum_{n \geq 0} \frac{f_n(y)}{n!} x^n$, where

$$f_n(y) = a_0 + e^y(b_0 + b_1 y + b_2 y^2) + e^{2y}(c_0 + c_1 y + c_2 y^2) + e^{3y}(d_0 + d_1 y + d_2 y^2 + d_3 y^3)$$

for $n \geq 5$ with

$$a_0(n) = 5 \cdot 3^{n-1}(n-1)n, \quad b_0(n) = 5 \cdot 3^n(n^2 - 9n - 24) + 5 \cdot 2^n(3n^2 + 15n + 24),$$

$$b_1(n) = -20 \cdot 3^n(n-6) - 30 \cdot 2^n(n+4), \quad b_2(n) = 15(3^n - 2^n)$$

$$c_0(n) = 15 \cdot 2^n(n^2 + 5n + 12) - 30(2n^3 + 7n + 6),$$

$$c_1(n) = 60(2n^2 + 2n + 3) - 15 \cdot 2^{n+2}(n+3), \quad c_2(n) = 15 \cdot 2^{n+2} - 60(n+1),$$

$$d_0(n) = -10(2n^3 + 7n + 6), \quad d_1(n) = 30(2n^2 + 2n + 3),$$

$$d_2(n) = -60(n+1), \quad d_3(n) = 20.$$

We always have $a_0, b_2, c_0, c_2, d_1, d_3 > 0$ and $b_1, c_1, d_0, d_2 < 0$, whereas $b_0 > 0$ only for $n \geq 11$. For $n \geq 11$ we therefore have $f_n(y) \geq a_0 + d_0 - d_1 + d_2 - d_3 > 0$, the dominating term being a_0 . For $9 \leq n \leq 10$ we get $f_n(y) \geq a_0 + b_0 + d_0 - d_1 + d_2 - d_3 > 0$.

The inequalities $f_n(y) \geq 0$ for $0 \leq n \leq 8$ can be proved by Taylor expanding $e^{3t} f_n(-t)$ and checking that the coefficients are nonnegative.

4.3.4 Positivity of $P_2(x, y)$

We have

$$\begin{aligned} \frac{1}{3}P_2(x, y) = & 10e^{2(x+y)} \left(x^2 + x(3-2y) + y^2 - 3y + 3 \right) \\ & + 10e^{2x+y} \left(4x^2 - 4x(y-3) - 7y + 10 \right) \\ & + 10e^{2x} \left(x^2 + 3x + 2 \right) + 5e^{3x} \left(3x^2 - 2x - 4 \right) \\ & - 10e^{x+2y} \left(2x^3 + x^2(6-4y) \right. \\ & \left. + x \left(2y^2 - 8y + 9 \right) + 2 \left(y^2 - 3y + 3 \right) \right) \\ & - 10e^{x+y} \left(2x^3 - 2x^2(y-3) + x(9-4y) - 3y + 6 \right) \\ & + 2e^{2y} \left(7x^4 + x^3(20-14y) + x^2 \left(7y^2 - 30y + 30 \right) \right. \\ & \left. + 10x \left(y^2 - 3y + 3 \right) + 5 \left(y^2 - 3y + 3 \right) \right) \\ & + 5e^{3x+y} (3x-2y+2)(x-4). \end{aligned}$$

We Taylor expand $P_2(x, y)$ in the form $\sum_{n \geq 0} \frac{f_n(y)}{n!} x^n$, where

$$f_n(y) = a_0 + e^y(b_0 + b_1y) + e^{2y}(c_0 + c_1y + c_2y^2), \quad n \geq 5,$$

with

$$a_0(n) = 15 \cdot 2^{n-1} (n^2 + 5n + 8) + 5 \cdot 3^n (n^2 - 3n - 12),$$

$$b_0(n) = -30 (2n^3 + 7n + 6) + 15 \cdot 2^{n+1} (n^2 + 5n + 10) + 5 \cdot 3^n (n^2 - 11n - 24)$$

$$b_1(n) = -10 \cdot 3^n (n - 12) - 15 \cdot 2^{n+1} (2n + 7) + 30(2n^2 + 2n + 3)$$

$$c_0(n) = 15 \cdot 2^{n-1} (n^2 + 5n + 12) - 30 (2n^3 + 7n + 6),$$

$$c_1(n) = 60 (2n^2 + 2n + 3) - 15 \cdot 2^{n+1} (n + 3), \quad c_2(n) = 15 \cdot 2^{n+1} - 60(n + 1).$$

We always have $a_0, c_0, c_2 > 0$ and $c_1 < 0$, whereas $b_0 > 0$ only if $n \geq 13$ and $b_1 < 0$ only if $n \geq 11$. For $n \geq 13$ we have $f_n(y) \geq 0$ since all the coefficients have correct sign. For $7 \leq n \leq 12$ we have $f_n(y) \geq a_0 - |b_0| - |b_1| > 0$.

When $n = 0, 1$, $f_n = 0$, while for $2 \leq n \leq 6$ it is enough to Taylor expand $e^{2t} f_n(-t)$ and check that the coefficients are nonnegative.

4.3.5 Positivity of $P_3(x, y)$

We have

$$\begin{aligned} P_3(x, y) = & 15e^{3x+y} \left(x^2 - 4x + 2y - 2 \right) \\ & + 30e^{2x+y} \left(2x^2 - 2x(y-3) - 3y + 5 \right) \\ & + 30e^{2x} \left(2x^2 + 6x + 5 \right) + 15e^{3x} \left(3x^2 - 4x - 6 \right) \\ & - 30e^{x+y} \left(2x^3 - 2x^2(y-3) + x(9-4y) - 3y + 6 \right) \\ & - 10e^x \left(2x^3 + 6x^2 + 9x + 6 \right) \\ & + e^y \left(28x^4 + x^3(80-28y) - 60x^2(y-2) - 60x(y-2) - 30(y-2) \right), \end{aligned}$$

where

$$f_n(y) = a_0 + e^y(b_0 + b_1y), \quad n \geq 5.$$

with

$$\begin{aligned} a_0(n) &= -10 \left(2n^3 + 7n + 6 \right) + 15 \cdot 2^n \left(n^2 + 5n + 10 \right) + 5 \cdot 3^n \left(n^2 - 5n - 18 \right), \\ b_0(n) &= -30 \left(2n^3 + 7n + 6 \right) + 15 \cdot 2^n \left(n^2 + 5n + 10 \right) + 5 \cdot 3^{n-1} \left(n^2 - 13n - 18 \right) \\ b_1(n) &= 30 \left(2n^2 + 2n + 3 \right) - 15 \cdot 2^{n+1}(n+3) + 10 \cdot 3^{n+1}. \end{aligned}$$

We always have $a_0, b_1 > 0$ whereas $b_0 > 0$ only for $n \geq 14$. For $n \geq 14$ we get $f_n(y) \geq a_0 - b_1 > 0$, the dominating term being $5 \cdot 3^n n^2$ from a_0 . For $6 \leq n \leq 13$ we write $f_n(y) \geq a_0 - |b_0| - |b_1| > 0$.

When $0 \leq n \leq 5$ it is enough to Taylor expand $e^t f_n(-t)$ and check that the coefficients are nonnegative.

5 Concluding Remark

Let us mention that the n -dimensional version of this problem, lower bounds on the entropy among all log-concave random vectors with a fixed covariance structure has a well known connection to Bourgain's slicing problem, a longstanding open question in convex geometry, which was resolved² by Klartag and Lehec [18], using new bounds, due to Guan [19], on the average trace of an isotropic random vector's covariance matrix as it undergoes "Stochastic Localization", see [20]. See Bizeul [21] for another proof and [22–24] for background. For a log-concave random variable X in $\mathbb{R}^n$, its

² Even more generally and recently, Klartag and Lehec have resolved the stronger "thin shell conjecture" see [17], known to imply the slicing conjecture.

isotropic constant can be defined as in [25, Equation 24] by

$$L_X := \left(\frac{\det \operatorname{cov}(X)}{e^{2h(X)}} \right)^{\frac{1}{2n}}.$$

The resolution of Bourgain’s slicing problem gives upper bound on L_X independent of dimension. In this formulation, Theorem 1.1 says that when $n = 1$, $L_X \leq \frac{1}{e}$ with equality when X is an exponential distribution. This gives an affirmative resolution, in dimension $n = 1$ case, of the so-called “strong slicing conjecture”, see Fradelizi and Sola [26], that the isotropic constant of a log-concave random variable in $\mathbb{R}^n$ attains its maximum at the product of exponential distributions. Building on the work of Klartag [27], [26] shows that the strong slicing conjecture implies the functional version of the Mahler conjecture. Here, the Mahler conjecture is the longstanding conjecture in convex geometry that the volume product of a convex body and its polar is minimized for the simplex, see [28] for further background. Unfortunately, the tools developed here are not tractable outside of dimension one and do not seem useful toward bounding L_X in general.

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